Cloud Courier Evidence and Reflection

Test context

> - **Device:** Samsung SM S731U1
> - **Platform:** Android 17 (API 37)
> - **Notification permission:** authorized
> - **FCM token:** current installation token was available and copied without being displayed or recorded
> - **Valid message:** “Campus event update” / “View the latest campus information”
> - **Valid custom data:** `action=update_ui`, `asset=promo`
> - **T4 change:** `asset=unknown`; all other fields remained the same

Results

Test	Expected handler/result	Observed result	Status	Evidence
T1 Foreground	`onMessage`; promo title/image	`onMessage`, accepted diagnostic, Student Center image, receipt 1	Pass	`T1_foreground.png`
T2 Background tap	`onMessageOpenedApp`; promo title/image	`onMessageOpenedApp`, accepted diagnostic, Student Center image, receipt 2	Pass	`T2_background_tap.png`
T3 Cold-start tap	`getInitialMessage`; promo title/image	`getInitialMessage`, accepted diagnostic, Student Center image, receipt 1 on the new launch	Pass	`T3_cold_start_tap.png`
T4 Invalid foreground	`onMessage`; reject unknown asset and preserve content	`onMessage`, rejection diagnostic, Student Center title/image preserved, receipt 4, no crash	Pass	`T4_invalid_payload.png`

Graduate reflection

Setup and payload

I registered a new Android Firebase app with the exact package name `com.mustafadev.cloud_courier` instead of reusing an Activity 13 registration. FlutterFire generated matching options, and initialization is awaited before listeners or widgets start. The `action` key must equal `update_ui`; the `asset` key is then mapped only to `default` or `promo`. This allowlist prevents a cloud value from becoming an arbitrary local asset path.

Lifecycle and evidence

T1 reached `onMessage`, T2 reached `onMessageOpenedApp`, and the notification-triggered T3 launch reached `getInitialMessage`, as the labeled screenshots show. `onBackgroundMessage` is separately registered for non-UI background work because its isolate cannot directly mutate widget state. These results establish behavior for one Samsung device, Android 17, and notification-plus-data payload; they do not guarantee delivery across

devices, force-stop conditions, connectivity states, or data-only payload policies. Permission authorizes notification display but does not prove delivery.

Validation and next step

T4 changed `asset` to `unknown`. The validator reported the error while preserving the prior title and Student Center image. Listener cancellation and mounted/disposed guards prevent callbacks from updating dead UI. An alternative would construct asset paths dynamically, but that is shorter and unsafe. A feasible next improvement is a retryable initialization status; success would be observable when a simulated token failure displays an error and a retry later enables secure token copying without duplicate listeners.

Submission checklist

- **Repository:** https://github.com/mustafadev123/in_class_14
- **Evidence:** four PNG files in `evidence/`
- **Reflection:** this document
- **Limitation:** results apply only to the tested device/platform/payload conditions described above